AI-Powered Detection of Deforestation Drivers - Solafune Competition

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Abstract

Deforestation is a critical environmental issue that leads to biodiversity loss and accelerates climate change [1]. Identifying the underlying drivers of deforestation is essential for developing effective forest management strategies. Manual inspection of satellite imagery is extremely time-consuming, especially given the continuous and large-scale nature of deforestation. This paper presents a machine learning-based segmentation approach that combines three distinct methods from peer-reviewed research to enhance model accuracy and robustness. Our experiments show that Object-Based Augmentation improved test performance by 6.7% over the baseline implementation, some methods promoting invariance achieved higher validation scores but struggled to generalize. These findings highlight the challenges of designing robust augmentation pipelines for real-world deforestation monitoring and provide insights for future conservation efforts.

1 Introduction

Deforestation remains a major environmental challenge, contributing significantly to biodiversity loss and climate change. The Solafune competition focused on identifying four major drivers of deforestation—grassland expansion, mining, logging, and plantations—using satellite imagery [2]. Recognizing and distinguishing between these land-use changes is critical for implementing targeted conservation actions. Despite national efforts, deforestation in critical areas such as the Amazon continues to rise [3], highlighting the need for improved monitoring tools.

In this work, we adopt and extend a publicly available baseline solution developed for the Solafune competition [4], making several modifications to improve its segmentation capabilities. Building on this foundation, we integrate three peer-reviewed machine learning techniques into the model to further enhance robustness and generalization. The novelty of our approach lies both in improving the baseline and in selecting, combining, and evaluating augmentation and regularization methods tailored for deforestation driver identification. We directly compare the individual and combined effects of these techniques to determine the most effective strategy for enhancing segmentation performance in a remote sensing context.

Our main contributions are (1) adapting and improving an open-source baseline for deforestation driver segmentation, (2) integrating three peer-reviewed augmentation and regularization methods, and (3) providing a comparative evaluation of their individual and combined effects on model generalization.

2 Method

Our approach builds upon a strong segmentation baseline and integrates two augmentation-based and one invariance-based techniques from recent literature to enhance model robustness and generalization. In this section, we describe the baseline model architecture, followed by detailed explanations of the three augmentation methods and how they were adapted and implemented in our pipeline.

The three methods are built upon the baseline method referenced on the website [4].

We proposed applying three methods: Object Detection Augmentation [5], Automatic Data Augmentation via Invariance-Constrained Learning [6], and Interpolation Robustness [7]. We believed that these approaches complemented one another, collectively enhancing the robustness and accuracy of segmentation for identifying deforestation drivers. Our premise, which we investigated in this paper, was that robustness and invariance are critical concerns in segmentation tasks involving geographical data, particularly when using augmentation strategies. We believed that Automatic Data Augmentation via Invariance-Constrained Learning and Interpolation Robustness could help regularize training, while more extensive augmentation methods such as Object Detection Augmentation (OBA) might contribute to overall improved accuracy of our baseline.

2.1 Baseline

The baseline model was built on a U-Net architecture [8] with an EfficientNetV2-S encoder [9], pre-trained on ImageNet [10], and implemented using Pytorch. It accepted 12-band Sentinel-2 satellite imagery [11] as input and produced segmentation masks for the

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four deforestation driver classes: plantations, grassland shrubland, mining, and logging.

The loss function we call L_{cls} combined Dice Loss [12] and Binary Cross-Entropy with Logits Loss [13], promoting both pixel-wise accuracy and region-level overlap with the ground truth.

While Dice Loss focused on region overlap, Binary Cross-Entropy encourages precise pixel-wise classification, offering a balanced loss function.

We used AdamW [14] as the optimizer, chosen for its stability and built-in regularization via decoupled weight decay. A cosine annealing scheduler with warmup [15] was applied to adjust the learning rate dynamically during training, allowing for better convergence and exploration of the loss landscape.

To improve spatial and lighting variance during training, we applied the augmentations listed in Table 1 to the input.

Augmentation	(p)	Description
ShiftScaleRotate	0.5	Randomly shifts, scales, and rotates the image. Useful for spatial variance.
RandomCrop (512x512)	1.0	Crops a 512x512 patch from the image. Always applied.
HorizontalFlip	0.5	Flips the image horizontally.
VerticalFlip	0.5	Flips the image vertically.
Transpose	0.5	Swaps height and width dimensions (diagonal flip).
RandomRotate90	0.5	Rotates the image by 90, 180, or 270 degrees.
RandomBrightness Contrast	0.2	Randomly changes brightness and contrast. Simulates lighting variations.

Table 1. Data augmentations used for the baseline solution.

The same baseline configuration was used across all experiments involving combinations of the proposed research papers. Table A.1 summarizes the hyperparameters used for training.

2.2 Object-Based Augmentation (OBA)

Object-Based Augmentation (OBA) is a data augmentation technique originally introduced by [5] to enhance object-level diversity in datasets with limited samples. Instead of applying global image transformations, OBA focuses on identifying and extracting foreground objects from segmentation masks and pasting them onto new or existing backgrounds. While initially developed for synthetic house generation, the method is generalizable to other domains. In the context of deforestation segmentation, OBA enables the model to learn a broader distribution of land-use patterns—such as plantations, grassland, and mining sites—by increasing their visual diversity and frequency in the training data.

In our implementation, we extracted polygon-defined objects from labelled training masks and overlay them onto selected background regions. These backgrounds are either sourced from the same image or from a separate dataset, controlled by a hyperparameter. Although our experiments used extracted objects placed onto backgrounds from the same image, we illustrated in Appendix (Figure A.3) how the pipeline can also support pasting objects onto backgrounds sourced from a separate dataset. This functionality was implemented but not applied during the final training runs. To ensure semantic plausibility, we constrained pasted objects to avoid overlapping with existing segmented regions. This helped maintain spatial consistency and avoids unrealistic occlusions.

A visual example of this process in our implementation is provided in Fig. A.2.

We applied OBA offline during preprocessing via our training dataloader, introducing synthetic images into each batch with a fixed probability. The final training set is therefore a mixture of real and augmented samples. This augmentation strategy substantially increased the variety of training examples, which we hypothesized would improve the model's generalization to under-represented or visually ambiguous deforestation classes.

To ensure consistency and facilitate reproducibility, we summarize the key hyperparameters used for our implementation of the OBA pipeline in Table A.2.

For implementation details and theoretical background, we refer the reader to the original paper by [5].

2.3 Automatic Data Augmentation via Invariance-Constrained Learning

One of the issues with applying augmentations is the difficulty in knowing when, which and how often augmentations should be applied. Wrongful usage can lead to biases that outweigh the benefits of augmenting. The paper "Automatic data augmentation via invariance-constrained learning" attempts to fix this issue [6]. By framing augmentation as an invariance-constrained learning problem and using Monte Carlo Markov Chain (MCMC) sampling, the algorithm decides if, and when to apply transformations [16].

In our implementation, we used the same augmentations as the baseline shown in table 1. However, instead of pre-sampling augmentations within the dataloader, we sample augmentations on a per-batch basis. This approach allowed us to move away from hand-crafted or fixed augmentation policies. Instead, augmentations was selected dynamically based on how well they preserved semantic consistency under the model's current parameters. We used the Metropolis-Hastings (MH) [17] variant of MCMC to sample transformations from a uniform proposal distribution, evaluating whether to accept each proposed augmentation based on the associated loss.

The hyperparameters specific for the invariance-constrained learning (ICL) method are highlighted in table A.3. The number of samples per input was set to 3 to closely match the level of augmentation applied in the baseline model.

The invariance-constrained learning (ICL) algorithm continuously adjusted the augmentations based on how strong the model performed on each augmentation. First, for each training sample, $m_{samples}$ augmentations from the set of augmentations G was chosen randomly. The set G is equal to table 1, with the exception of RandomCrop which was always applied in the dataloader. Each candidate was evaluated based on how much it increased the model's loss. Using the Metropolis-Hastings criterion, three augmentations was sampled, favouring transformations that produced higher losses, highlighting areas where the model was not yet invariant. The probabilistic nature of MCMC means that there is a non-zero chance that an augmentation with lower loss is accepted. This smooths out the augmentation distribution and promotes exploration of different augmentations instead of always selecting the worst-case transformation.

The model was then updated using a Lagrangian objective that balanced the original loss with the constraint violation, weighted by a dual variable γ [18]. This variable was updated iteratively to ensure the constraint is met. A larger γ will increase the penalty of performing poorly on an augmentation. Through this primal-dual optimization loop, the algorithm adjusted the augmentation policy. Early stopping was implemented, to help avoid overfitting.

2.4 Interpolation Robustness

Interpolation Robustness [7], a novel approach to addressing domain shift invariance, was applied to the problem at hand. The authors of Interpolation Robustness proposed viewing domain shifts as an interpolation challenge across domains. While the method was primarily demonstrated in the context of classification, it was described as model-agnostic. In light of this, we extended its core principles to our segmentation task for detecting deforestation drivers. We aimed for incorporating [7] to serve as an effective regularization technique, enhancing the model's robustness to variations in spectral bands and potentially other domains.

In our adaptation of Interpolation Robustness, we encoded two images from different domains, interpolated between their UNet bottleneck representations [8], and created a new latent image z from the learned function T. We decoded this interpolated representation and calculated L_{int}, inspired by Equation (4) from the original paper. Our L_{cls} was modified to follow the baseline loss function described earlier, while the ℓ_2 loss remained unchanged.

$$L_{int} = L_{cls} + ||Z(x, x'; \phi, 1, \psi) - E(x')||$$

In addition to L_{int}, we calculated the mean L_{cls} of the two domain images, giving the final loss function L, where λ was the regularization parameter:

$$L = \frac{L_{\mathrm{cls}_{d1}} + L_{\mathrm{cls}_{d2}}}{2} + \lambda L_{int}$$

This approach was intended to encourage the model to learn domain-invariant features across different domains or spectral bands, rather than overfitting to domain-specific characteristics. While spectral band interpolation alone may yield limited gains, the method was expected to generalize better to more meaningful domain shifts—such as temporal changes—and could be particularly relevant in our case for handling augmentation-induced shifts like the background replacement in OBA.

3 Results

3.1 Training Results

During the training of our baseline model—prior to integrating any research paper implementations—we obtained the results summarized in Table 2.

Best Epoch	Train Loss	Val Loss	Val F1
0	1.310	1.360	0.081
1	1.310	1.200	0.136
10	0.928	0.861	0.319
20	0.857	0.799	0.390
29	0.835	0.781	0.425

Table 2. Baseline training and validation metrics for selected epochs based on the new training output.

Each research paper method was trained using the same baseline architecture. The best epoch, along with the corresponding training and validation metrics for each method, is summarized in Table 3. Interpolation Robustness (IR) and Invariance-Constrained Learning (ICL) were not evaluated together, as they are intended to complement the OBA pipeline rather than each other.

Method	Epoch	Train Loss	Val Loss	Val F1
OBA	28	0.761	0.795	0.460
ICL	12	1.871	0.825	0.658
IR	19	0.578	1.150	0.465
OBA+ICL	11	1.916	0.834	0.701
OBA+IR	12	0.66	0.465	0.33

Table 3. Training and validation metrics for the best epoch of each method.

3.2 Evaluation on unseen data

Method	Epoch	Val F1	Test F1
Baseline	29	0.425	0.508
OBA	28	0.460	0.542 (+6.7%)
ICL	12	0.658	0.445 (-6.3%)
IR	–	–	–
OBA+ICL	11	0.701	0.484 (-2.4%)
OBA+IR	–	–	–

Table 4. Comparison of segmentation performance across methods for the best epoch of each method. Validation F1 represents the best score achieved during training. Test F1 is based on the Solafune public leaderboard.

The baseline model achieved a test F1 score of 0.508. Applying OBA increased the test F1 to 0.542. The ICL model, despite achieving a validation F1 of 0.658, obtained a test F1 of 0.445.

The combination of OBA and ICL reached a validation F1 of 0.701 but a test F1 of 0.484.

Interpolation Robustness (IR) produced no meaningful results on unseen data. Most predicted segments fell below the minimum area threshold and were filtered out, indicating that the model output primarily consisted of noise.

When combined with Object-Based Augmentation (OBA), Interpolation Robustness also yielded no results, This suggests that IR applied but, destabilizes training and would require more carefully initialization, warm-up and other general optimizations.

4 Discussion

The baseline model demonstrated the ability to learn basic deforestation patterns, achieving a validation F1 score of 0.425 and a test F1 score of 0.508. However, its performance remained limited, likely due to the challenges posed by class imbalance and the complexity of distinguishing similar land-use types (Appendix, Figure A.1).

Applying Object-Based Augmentation (OBA) led to a notable improvement in test F1 score (+6.7% over baseline). This suggests that synthetic object-level augmentation effectively increased visual diversity in the training set, helping the model generalize better to unseen imagery.

In contrast, the Invariance-Constrained Learning (ICL) approach achieved the highest validation F1 score (0.658) but underperformed on the test set, dropping 6.3% compared to the baseline. This suggests that ICL promoted robustness during validation but led to overfitting on augmentation patterns that did not generalize to unseen data. Despite using early stopping, the discrepancy points to a distribution mismatch between training/validation and real-world data. The significant class imbalance (Figure A.1) may have further amplified this effect, with learned invariances that might not generalize equally across all classes. Additionally, the small batch size (4 vs. 128 in the original paper) likely introduced instability in estimating invariance constraints. Since the ICL method relies on batch-level statistics for primal and dual updates, too few samples can produce noisy gradient estimates, resulting in less stable and less effective optimization.

The combination of OBA and ICL further improved validation F1 to 0.701, but again resulted in a test performance drop (-2.4% compared to baseline). This highlights the risk of relying solely on validation metrics without proper testing on unseen data, particularly when complex augmentation strategies are involved.

Interpolation Robustness (IR) performed poorly on the validation data. A likely cause is that the parametric interpolation introduced excessive noise early in training, destabilizing learning and preventing the model from developing useful representations. As a result, we were unable to produce a valid submission that could be evaluated on the test data. Unlike Invariance-Constrained Learning (ICL), which at least achieved strong validation performance before failing on test data, IR showed no strong performance at any stage. Additionally, since the Solafune competition evaluates performance solely based on IoU rather than domain generalization, we could not fully assess the benefits IR was originally designed to provide.

Our best performing approch was the OBA method with a test F1 score of 0.542. The winner of the Solafune contest reached a test F1 score of 0.721. This suggests that while we saw some improvements over the baseline, further work is needed to achieve state-of-the-art performance.

Overall, our findings emphasize that achieving high validation scores alone does not guarantee strong generalization. Careful design of augmentation and robustness techniques, along with thorough testing on independent datasets, is crucial for deploying reliable segmentation models in real-world scenarios.

4.1 Future Work

Several avenues could further improve the segmentation model presented in this study. First, replacing the current EfficientNetV2-S encoder [9] with a more powerful and domain-specific backbone—particularly one pretrained on remote sensing imagery—may lead to better feature representations.

Second, expanding and diversifying the training dataset, for instance by collecting additional Sentinel-2 imagery and annotating underrepresented classes, could significantly enhance generalization to new regions and land-use types.

Interpolation robustness leaves much to be desired. A major hindrance to training for interpolation robustness is the immense computational overhead required to interpolate between images and their domains. This forced the use of small batch sizes, and it remains unclear whether larger batch sizes—if feasible—would improve training. Additionally, early in training, it would be beneficial to apply early stopping or dynamically adjust the regularization parameter λ on the interpolation loss L_{int}, reducing its weight when it is less effective in the initial stages. Incorporating a Leave-One-Out accuracy metric would better capture the intent of interpolation robustness, by directly evaluating the model's ability to generalize and perform on unseen domains.

Finally, model performance may benefit from systematic hyperparameter optimization, such as grid search or Bayesian optimization, to better explore the configuration space, along with increasing the number of epochs to allow models sufficient time to converge.

5 Conclusion

In this work, we developed a segmentation model to identify drivers of deforestation using Sentinel-2 satellite imagery. Building upon a U-Net baseline with an EfficientNetV2-S encoder, we explored the impact of three techniques—Object-Based Augmentation (OBA), Invariance-Constrained Learning (ICL), and Interpolation Robustness (IR)—on model performance.

Our results demonstrated that Object-Based Augmentation improved test generalization by 6.7%, confirming the importance of enhancing object-level diversity in training data. In contrast, methods promoting invariance, such as ICL and IR, achieved higher validation scores but failed to generalize well to unseen data. This highlights a key challenge: techniques that improve validation robustness may not always transfer effectively to real-world conditions, particularly in highly imbalanced and noisy datasets like remote sensing imagery.

Overall, our findings emphasize that building robust deforestation detection models requires not only strong validation performance, but also careful evaluation on independent, unseen datasets. Future research should focus on developing augmentation and regularization methods that balance invariance with real-world generalization, as well as addressing issues like class imbalance and domain shifts more explicitly.

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A Appendix

Hyperparameter	Value / Description
Random Seed	42
Epochs	30
Batch Size (train/val/test)	4
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-2}
Min Learning Rate	0.0
Warmup Learning Rate	1×10^{-5}
Prediction Threshold	0.5
Post-processing Area Threshold	20,000 pixels (minimum segmented area to retain)

Table A.1. Hyperparameters used in baseline training across all experiments.

Hyperparameter	Value / Description
OBA Probability	0.6 — Chance of applying OBA to each image during training
Objects per Image	8 — Number of objects to paste per sample
Max Extraction Attempts	10 — Tries per object to find a valid extraction
Extract from Same Image	True — Separate dataset for backgrounds currently not used in our setup. If set to False, probability of using a separate set of images for background is controlled by `BACKGROUND_PROB` = 0.3

Table A.2. OBA-specific hyperparameters used during training.

Hyperparameter	Value / Description
Epsilon (ϵ)	0.05 — Threshold for the invariance constraint slack
Primal Learning Rate (η_p)	0.01 — Step size for model parameter updates
Dual Learning Rate (η_d)	0.01 — Step size for dual variable updates
Initial Dual Variable (γ)	0.5 — Starting value for Lagrange multiplier
MH Sampling Steps (n_{MH})	2 — Number of Metropolis-Hastings iterations
Samples per Input (m_{samples})	3 — Number of accepted augmentations used per sample

Table A.3. Invariance-constrained learning hyperparameters used during training.

Figure A.2. Effect of Object-Based Augmentation (OBA) on training data. The upper panels shows original satellite imagery, while the lower panels shows the same scene with OBA-applied synthetic objects overlaid. This demonstrates how OBA introduces new spatial configurations of deforestation drivers, improving visual diversity.

Figure A.1. Class distribution in the training set. This histogram shows the frequency of each class label across all annotated training images, useful for identifying class imbalance.

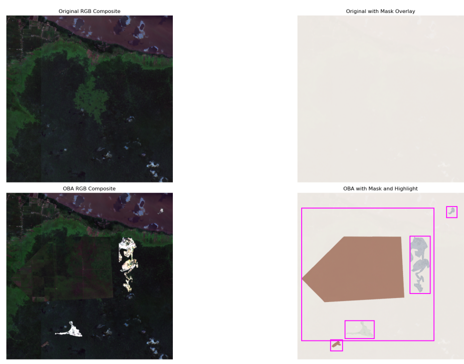

Figure A.3. Illustration of Object-Based Augmentation (OBA) using a separate background dataset. Foreground objects extracted from annotated images are overlaid onto backgrounds sourced from a different dataset. This process increases diversity by creating new spatial compositions of deforestation drivers without relying solely on original imagery, enhancing the model's exposure to varied land-use patterns during training.

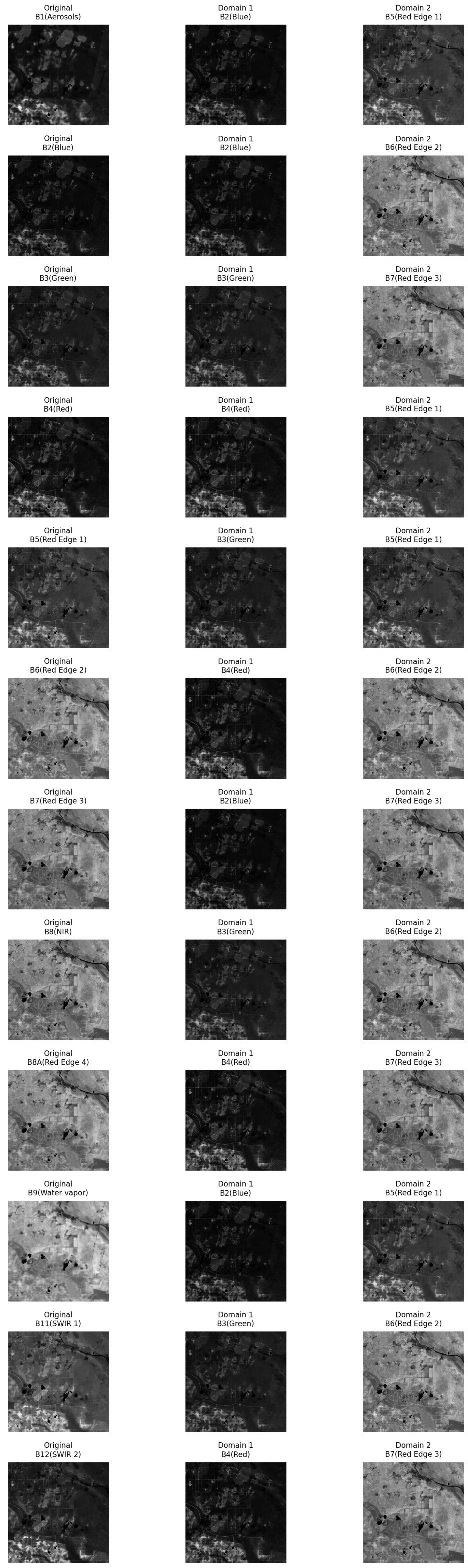

Figure A.4. Output of the `domain_image_split` function. Each band of a 12-band TIFF image is padded by repetition to create domain-specific images. Left: original 12-band image with all domains. Middle: image with only RGB bands. Right: image with only Red Edge bands.